Deep Learning-Based Estimation of Emission Time and Arrival Time in Diffusive Multi-Receiver Molecular Communication

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***Abstract*—Diffusive molecular communication (DMC) utilizes the emission, diffusion and reception of molecules to transmit information. It has promising prospects in the field of drug delivery. The estimation of emission time and arrival time of molecules in DMC system plays important roles in the resource consumption at the receivers. Existing traditional strategies for the derivation of emission time and arrival time mainly focus on known channel state information (CSI). In this paper, we propose a deep learning method for estimating emission time and arrival time of the molecules in DMC system with unknown CSI by using Transformer-based model, respectively. The simulation results show that the emission time and arrival time of molecules can be accurately estimated by the Transformer-based model which exhibits better estimation and generalization abilities than deep neural network (DNN) model.**

***Index Terms*—Diffusive molecular communication, channel state information, emission time, arrival time, transformer-based.**

I. Introduction

MOLECULAR communication (MC) defines information transmission which relies on the emission, diffusion and reception of molecules within a medium [1]. The transmitter encodes information by releasing specific chemical molecules, which diffuse through the medium and are ultimately detected and decoded by the receiver [2]. The Nano Internet of Things (IoNT) promotes the development of MC, which finds applications in health monitoring, environmental sensing, and other domains [3]. In the medical field, the nanomachines can monitor the health status of organisms and exchange data with other nano-devices. Additionally, diffusive molecular communication (DMC) enables efficient internal communication within organisms by transmitting information through information molecules in fluid environments [4], [5].

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In the microenvironment, the limited processing capacity and storage resources of nanomachines necessitate the optimization of the communication process of DMC system. However, the performance of this optimization is influenced by several factors, such as the number of released molecules, the emission time, the arrival time and the detection time. In order to improve the system performance, Rudsari et al. [6] developed a dynamic time-division multiple access drug release mechanism that optimized both the number of molecules released by the transmitter and the duration of detection interval at the receiver, thereby minimized the bit error rate (BER) for each transmitter. Recently, estimating the time of molecules released by the transmitter and the arrival time as well as the detection time interval has emerged as a significant topic. The emission time is very important to the receiver. The authors [7] approached the emission time by estimating the non-zero emission duration of molecules using a rectangular input concentration and the maximum likelihood estimation method. When the emission time was unknown, it complicated the problem of precise location of nanomachines. The localization method was proposed to achieve comparable individual localization performance to the impulse concentration based on unknown emission time [8].

Some researchers have examined the arrival time of molecules at the receiver in DMC environment. Haselmayr et al. [9] utilized a moment matching approximation with a normal inverse Gaussian distribution in a one-dimensional (1D) flow induced diffusion channel to estimate the arrival time for molecules to reach the receiver. The work [10] derived the mathematical expressions for the probability density function of the first hitting time which was the arrival time of molecules in 1D diffusive mobile molecular communication (MMC). The authors [11] assumed the first arrival time for different molecules was correlated and then analyzed the synchronization performance based on the maximum likelihood estimator. Some work studied the detection time in DMC system. Akdeniz et al. [12] introduced the shift-π method, which reduced the BER in the DMC channel by delaying the time at which the receiver absorbed the molecules. In 1D and three-dimensional (3D) DMC systems where the exact location of the interference source was an unknown receiver, Cao et al. [13] significantly reduced the BER in the channel by optimizing the optimal detection interval at the receiver. Cheng et al. [14] employed a multi-objective optimization algorithm to determine the

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optimal detection time for the receiver in a 3D multi-user MMC environment.

Recently, there have been growing interests in incorporating deep learning functionalities into DMC systems. This integration also explored the potential benefits of employing energy-efficient algorithms for deep learning on low power devices [15]. Deep learning has found extensive applications in DMC. Methods for detection signals based on deep learning were studied. Cheng et al. [16] introduced a relay signal detector based on a Transformer model in a cooperative multi-hop DMC system, deriving higher signal detection accuracy under two transmission strategies including single-molecule type and multi-molecule type. Bai et al. [17] proposed an improved time-domain convolutional network (TCN) for signal detection in a DMC system using magnetotactic bacteria (MTB) as the information carrier. Torres Gómez et al. [18] explored the interpretability of neural networks in signal detection, employing the local interpretable model-agnostic explanation method for binary transmission in DMC channels. In a two-dimensional (2D) static DMC environment, Kose et al. [19] utilized a deep neural network (DNN) to locate silent entities without a clear analytical formula representing the received signal. Baydas et al. [20] introduced a supervised learning algorithm to estimate the number of molecules in the multiple-input multiple-output (MIMO) channels, utilizing convolutional neural network (CNN) technology to replace traditional threshold detection methods. Cheng et al. [21] derived the average BER for MIMO systems and employed DNN to optimize the number of molecules emitted by each transmitter, aiming to minimize the average BER of this system. Shrivastava et al. [22] proposed a neural network detector for a MMC environment, optimizing the network using the Broyden-Fletcher-Goldfarb-Shanno (BFGS) and Levenberg-Marquardt (LM) algorithms. Lu et al. [23] developed a signal detector based on a Transformer model, demonstrating superior performance even in higher noise conditions. Farsad and Goldsmith [24] proposed a sliding bidirectional recurrent neural network (SBRNN) for signal detection in the absence of channel state information (CSI). Sun and Wang [25] introduced an innovative recurrent neural network architecture which was designed to learn and estimate the transmitted signal sequences without prior knowledge of the channel model. Cheng et al. [26] investigated the impacts of neural networks on the signal detection in time-varying MMC system, proposing an Informer-based signal sequence detector that enhanced the system performance when the CSI was unknown.

In this paper, we use the deep learning method to estimate the emission time and the arrival time of molecules with unknown CSI. The proposed method can be utilized in the application scenarios, such as the location of the transmitter and the receiver in DMC. Specifically, for the location issues, how to locate the transmitter when the emission time is unknown and how to locate the receiver when the emission time is known needs to be studied. The motivations of this paper are based on the following considerations:

(1) The existing work assumed that the emission time of molecules is instantaneous and known, which means that the receiver can know the time when the transmitter releases molecules through clock synchronization. However, in some specific applications, the emission time is not known to the receiver, then the receiver can not determine the detection time which further complicates the performance analysis of this DMC system. Under such a case, how the receiver estimates the emission time is crucial for optimizing system performance and reducing resource consumption.

(2) The arrival time is defined by the time when the first molecule arrives at the receiver. Usually, the detection time of the receiver in DMC system starts when the transmitter releases molecules. Considering the fact that the molecules needs some time to arrive at the receiver due to the diffusion, the receiver can delay the detection time and starts detection after it receives the first molecule. Then this method can reduce resource consumption at the receiver. Therefore, when the emission time is known, predicting the arrival time of molecules is of great significance.

(3) Existing traditional methods for obtaining emission time and arrival time mainly focused on known CSI. However, when the CSI is unknown, these traditional strategies are not effective. How to estimate the emission time and arrival time of molecules in DMC with unknown CSI is a challenge work which needs to be studied. In this paper, we use the deep learning methods to solve these two problems with unknown CSI based on Transformer model.

Based on the above motivations, the main contributions of our paper are as follows:

(1) Considering the scenarios that one transmitter and multiple receivers in two dimensional DMC system, we propose a method for estimation of emission time and arrival time in DMC system with unknown CSI by using Transformer-based model. The Monte Carlo simulation is utilized to capture the positions of molecules released by the transmitter in the environment.

(2) Using the training and testing datasets generated from the reception of molecules at the receivers and the coordinates of the transmitter, the Transformer-based model is employed to solve the issues of estimating the emission time and the arrival time, respectively. The input and the output of the Transformer-based model when obtaining the emission time are different from those when estimating the arrival time of molecules. Then we analyze and evaluate the performance of this model.

(3) Simulation results show that the emission time and arrival time of the molecules can be estimated accurately by using Transformer-based model, respectively. The Transformer-based model can exhibit better abilities of estimating the emission time and arrival time. It demonstrates faster convergence and lower loss compared with the DNN-based model.

The rest of our article is arranged as follows. Section II provides a detailed introduction of the transmission and reception of molecules in the DMC environment. Sections III and IV summarize the data generation, the considered neural network model, the training and testing process for estimation of emission time and arrival time, respectively. In Section V, we conduct simulation experiments to compare the performance

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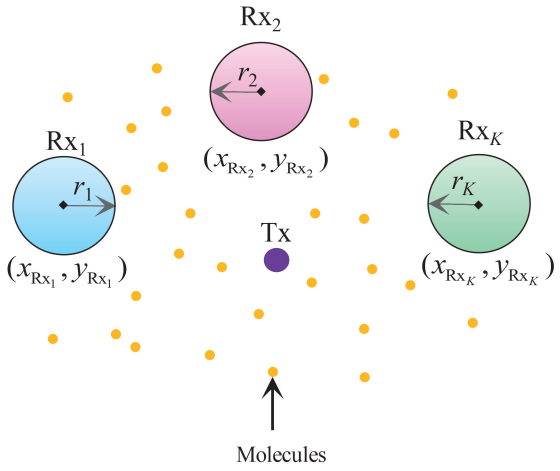


Fig. 1. Topology model.

differences between the Transformer-based model and the DNN-based model, and then we analyze the factors which have influenced the performance of the Transformer-based model. Section VI concludes our work.

II. System Model

We consider a DMC system consisting of a point transmitter (Tx) and K circular receivers $\mathrm{Rx}1$, $\mathrm{Rx}2$, …, and $\mathrm{Rx}K$. The radii of K receivers are r_1, r_2, …, and r_K, respectively. In Fig. 1, the model operates in a static 2D unbounded environment with no flow velocity. For this DMC system, our assumptions are as follows:

(1) In the 2D environment, the coordinate of Tx is $(x_{\mathrm{Tx}}, y_{\mathrm{Tx}})$, and the center coordinates of $\mathrm{Rx}1$, $\mathrm{Rx}2$,…, and $\mathrm{Rx}K$ are $(x_{\mathrm{Rx}1}, y_{\mathrm{Rx}1})$, $(x_{\mathrm{Rx}2}, y_{\mathrm{Rx}2})$,…, and $(x_{\mathrm{Rx}K}, y_{\mathrm{Rx}K})$, respectively. The number of molecules emitted by Tx is represented by N, and the diffusion coefficient of the molecules is denoted by D.

(2) Based on the existing research [27], [28], we assume that there is perfect time synchronization between the transmitter and all receivers. We use T to represent the detection time in the system and Δt is defined as each time step. In terms of selecting the value of Δt, the smaller value of Δt can more accurately simulate the movement of molecules and the situation that the receiver captures the molecules. When Δt becomes small enough, the sequence of the number of molecules captured by the receiver in the dataset will become much longer, which will lead to increase the complexity of the training by the model through experiments. Therefore, we divide T into M time steps. Then we have $\Delta t{=}T/M$. We assume that all receivers can count the number of molecules captured in each Δt.

In the DMC system, the diffusion of molecules obeys the normal distribution. Therefore, we use Δx_{mol} and Δy_{mol} to represent the displacement of molecules along the x-axis and y-axis in each Δt, respectively. Δx_{mol} and Δy_{mol} obey normal distribution:

$$\begin{aligned} \Delta x_{\mathrm{mol}} &\sim N(0, 2D\Delta t), \\ \Delta y_{\mathrm{mol}} &\sim N(0, 2D\Delta t). \end{aligned} \tag{1}$$

The calculation methods of Δx_{mol} and Δy_{mol} are

$$\begin{aligned} \Delta x_{\mathrm{mol}} &= \varepsilon_x \times \sqrt{2D\Delta t}, \\ \Delta y_{\mathrm{mol}} &= \varepsilon_y \times \sqrt{2D\Delta t}, \end{aligned} \tag{2}$$

where ε_x and ε_y represent random parameters following a normal distribution:

$$\begin{aligned} \varepsilon_x &\sim N(0, 1), \\ \varepsilon_y &\sim N(0, 1). \end{aligned} \tag{3}$$

We utilize $(x_{\mathrm{mol}}(t+\Delta t), y_{\mathrm{mol}}(t+\Delta t))$ to represent the position of the molecule at time $(t+\Delta t)$, $(x_{\mathrm{mol}}(t), y_{\mathrm{mol}}(t))$ is denoted by the position of the molecule at time t. Then the displacement of the molecule in the environment is calculated as follows:

$$\begin{aligned} x_{\mathrm{mol}}(t+\Delta t) &= x_{\mathrm{mol}}(t) + \Delta x_{\mathrm{mol}}, \\ y_{\mathrm{mol}}(t+\Delta t) &= y_{\mathrm{mol}}(t) + \Delta y_{\mathrm{mol}}. \end{aligned} \tag{4}$$

After the molecule is released by Tx, the position information of the molecule at time t can be obtained according to Eq. (4), and the distance between the molecule and each receiver can also be calculated. When the distance between the molecule and the k-th receiver is less than or equal to r_k, the molecule will be received by the k-th receiver. Then this molecule is removed from the environment. The distance between this molecule and the k-th receiver coordinates at time t is calculated as follows:

$$\sqrt{[x_{\mathrm{mol}}(t) - x_{\mathrm{Rx}_k}]^2 + [y_{\mathrm{mol}}(t) - y_{\mathrm{Rx}_k}]^2} \le r_k, \tag{5}$$

where r_k is the radius of the k-th receiver. $(x_{\mathrm{Rx}_k}, y_{\mathrm{Rx}_k})$ is the coordinate of the k-th receiver.

III. Transformer-Based Prediction of Emission Time

A. Data Generation

In the 2D environment, the emission time of the molecules is random. The transmitter instantaneously emits N molecules with the same type at a random time which is denoted by t_{emission}. t_{emission} is within $[0, T]$ and all molecules are of one type. The receiver can capture enough molecules to predict the emission time of the molecules before stopping the detection. To enhance the accuracy of emission time estimation, we employ the Monte Carlo [29] simulation experimental method to capture the positions of molecules in the environment.

It is worth noting that before the molecules are released, the receiver counts that the number of captured molecules in each Δt is 0. These numbers 0 can help the neural network to determine whether the molecule has been released.

In the simulation experiment, the coordinates of Tx in the 2D environment are completely random. When selecting the coordinate range of Tx, we should consider any possible position of Tx around each receiver. At the same time, we need to avoid the situations that Tx is too far from the receiver which will result in the loss of communication and the molecules cannot reach the receiver within $[0, T]$. Under such a case, the number of the molecules captured by the receivers within $[0, T]$ are all 0, then the sequence which is composed

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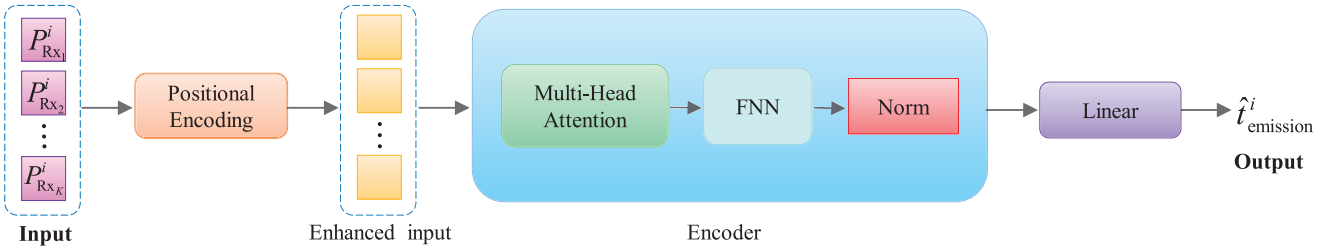


Fig. 2. Transformer-based model for estimating the emission time.

of the number of received molecules in the dataset cannot be used to train the model. Under the above requirement, we select a square coordinate region and generate the data in the coordinate range of Tx as follows:

$$\begin{aligned} -20\mu\text{m} \leq x_{\text{Tx}} \leq 20\mu\text{m}, \\ -20\mu\text{m} \leq y_{\text{Tx}} \leq 20\mu\text{m}. \end{aligned} \tag{6}$$

Considering the situation that the coordinates of Tx overlap with any receiver is impractical, we need to remove these coordinates, which cannot be used to generate the training data.

Starting $t = 0$, each receiver counts the number of captured molecules in each Δt until the end of time T. We use λ_{Rx_k} to denote the sequence of molecules captured by the k-th receiver in $[0, T]$, so we have

$$\lambda_{\text{Rx}_k} = \left[\lambda_{\text{Rx}_k}(\Delta t), \lambda_{\text{Rx}_k}(2\Delta t), \ldots, \lambda_{\text{Rx}_k}(M\Delta t)\right], \tag{7}$$

where $\lambda_{\text{Rx}_k}(\Delta t), \lambda_{\text{Rx}_k}(2\Delta t), \ldots, \lambda_{\text{Rx}_k}(M\Delta t)$ are denoted by the number of molecules absorbed by Rx_k during Δt, $2\Delta t, \ldots, M\Delta t$, respectively.

We use $P^i_{\text{Rx}_k}$ to represent the sequence of molecules captured by the k-th receiver in the i-th sample in the dataset, then the representation is

$$P^i_{\text{Rx}_k} = \left[\lambda^i_{\text{Rx}_k}(\Delta t), \lambda^i_{\text{Rx}_k}(2\Delta t), \ldots, \lambda^i_{\text{Rx}_k}(M\Delta t)\right]. \tag{8}$$

When the emission time t_{emission} of the molecules is too large, the molecules cannot arrive at any receiver in $[0, T]$. Therefore, we set the latest emission time t_{latest} ($t_{\text{latest}} < T$), and let t_{emission} satisfy the following conditions:

$$t_{\text{emission}} < t_{\text{latest}}. \tag{9}$$

Then the representation of the input of the i-th sample W^i_{emission} in the corresponding dataset is as follows:

$$W^i_{\text{emission}} = \left[P^i_{\text{Rx}_1}, P^i_{\text{Rx}_2}, P^i_{\text{Rx}_3}, \ldots, P^i_{\text{Rx}_K}\right]. \tag{10}$$

We assume that the t^i_{emission} is the emission time of the molecules in the i-th simulation experiment, which is the output of the i-th sample. We use C^i_{emission} to denote the i-th sample in the dataset, which is represented by

$$C^i_{\text{emission}} = (W^i_{\text{emission}}, t^i_{\text{emission}}). \tag{11}$$

Therefore, I_1 represents the number of samples in the dataset, then the dataset $F^{I_1}_{\text{emission}}$ is described as

$$F^{I_1}_{\text{emission}} = \left\{C^1_{\text{emission}}, C^2_{\text{emission}}, C^3_{\text{emission}}, \ldots, C^{I_1}_{\text{emission}}\right\}. \tag{12}$$

B. Considered Model

The self-attention mechanism is the core of the Transformer model, which achieves efficient representation learning by effectively capturing the dependencies between elements in the sequence. In order to predict the emission time of the molecules more accurately, we use the Transformer-based neural network model. Fig. 2 shows the Transformer-based model structure, which is used to predict the emission time of molecules. The inputs correspond to the sequence of molecules captured by K receivers in $[0, T]$. For the i-th sample, the input is $[P^i_{\text{Rx}_1}, P^i_{\text{Rx}_2}, P^i_{\text{Rx}_3}, \ldots, P^i_{\text{Rx}_K}]$. The details of the sequence have been introduced in Eq. (7) and Eq. (8). When processing the sequence data, the preservation of sequence information is critical. The Transformer model utilizes a self-attention mechanism to analyze the entire sequence, which can lead to the loss of positional information. To address this issue, positional encoding is introduced to provide each input vector with information regarding its position, enabling the model to comprehend both the relative and absolute positions of each element within the sequence. Specifically, we employ a fixed positional encoding scheme based on sine and cosine functions as follows:

$$\begin{aligned} \Phi_{(pos,2\sigma)} &= \sin(pos/10000^{2\sigma/\gamma}), \\ \Phi_{(pos,2\sigma+1)} &= \cos(pos/10000^{2\sigma/\gamma}), \end{aligned} \tag{13}$$

where $\Phi_{(pos,2\sigma)}$ denotes the even dimension using a sine function to calculate the encoding, and $\Phi_{(pos,2\sigma+1)}$ defines the odd dimension using a cosine function to calculate the encoding. 2σ represents the even index in the sequence of number of molecules captured by the receiver, and $(2\sigma + 1)$ is the odd index. γ is the dimension of the hidden layers.

After incorporating positional encoding into the input sequence, we employ a multi-head attention mechanism. This approach utilizes multiple attention heads to simultaneously compute the correlations among the elements in the input sequence, thereby extracting pertinent information. This method effectively captures the characteristics of the reception sequences by each receiver. The multi-head attention mechanism [30] can be described as follows:

$$\mathcal{M}(Q, G, V) = \text{Softmax}\left(\frac{QG^{\text{T}}}{\sqrt{d}}\right)V. \tag{14}$$

In Eq. (14), the query vector (Q) serves to retrieve the most relevant information regarding the emission time of the molecules from the input sequence, guiding the model to concentrate on the most pertinent aspects of the current task. The key vector (G) is compared with the query vector to

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compute their similarity, which assists the model in evaluating the correlations and dynamically selecting the relevant input information. This process enhances the model's understanding and generative capabilities. The value vector (V) encapsulates the specific information associated with the input. The attention mechanism ultimately generates the output by applying weights to the elements in value vector. The hyperparameter d plays a crucial role in controlling the dimensionality of the key vector and performing numerical scaling during the calculation of attention weights.

Additionally, in the feedforward neural network (FNN), the dropout layer is integrated with the residual connection, which helps the model to mitigate overfitting and to improve the model's feature representation capabilities. During the training of neural networks, normalization can accelerate the convergence of the optimization process, enhance the model's generalization ability, reduce the risk of overfitting and improve the model's robustness on unknown data. Finally, we use the linear layer to output the emission time $\hat{t}^{i}_{\text{emission}}$ of the molecules.

C. Training and Testing Process

Based on the simulation experiment range selected by formula (6), we take the point with a discrete step size of 1μm along the x-axis and the y-axis as the coordinates of Tx with a total of 1,600 coordinates. After removing the coordinates that do not meet the conditions, there are 1,560 coordinates. It is noted that we will continue to select different molecular emission time by simulation experiments to generate samples in each reasonable coordinate. For each reasonable coordinate in each experiment, t_{emission} randomly takes a value in $[0, t_{\text{latest}}]$ with Δt as the step size, so we finally get 124,800 samples in the dataset, with 80% allocated to the training dataset and 20% to the testing dataset. We use N_1 to represent the number of samples in the testing dataset and $N_1 = 24,960$. S_1 is a batch size and $S_1 = 64$. The learning rate is set to 0.005.

We choose Mean Squared Error (MSE) as the loss function in the training process. Assuming that u is the index of the u-th sample in S_1. t^{u}_{emission} is the real molecular emission time corresponding to the u-th sample. $\hat{t}^{u}_{\text{emission}}$ is the predicted molecular emission time corresponding to the u-th sample. The MSE loss function for estimating the emission time is calculated as follows:

$$\text{MSE}_{\text{emission}} = \frac{1}{S_1}\sum_{u=1}^{S_1}\left(t^{u}_{\text{emission}} - \hat{t}^{u}_{\text{emission}}\right)^2. \quad (15)$$

In the testing process, we use α_v to represent the prediction error of the v-th sample in the testing dataset. The calculation method is

$$\alpha_v = |t^{v}_{\text{emission}} - \hat{t}^{v}_{\text{emission}}|, \quad (16)$$

where t^{v}_{emission} is the real molecular emission time corresponding to the v-th sample. $\hat{t}^{v}_{\text{emission}}$ is the predicted molecular emission time corresponding to the v-th sample. We establish an error threshold to evaluate the difference between t^{v}_{emission} and $\hat{t}^{v}_{\text{emission}}$. N^{v}_{c} is used to evaluate the accuracy of each sample prediction as follows:

$$N^{v}_{c} = \begin{cases} 1, \alpha_v < \theta, \\ 0, \alpha_v \geq \theta, \end{cases} \quad (17)$$

where θ represents the error time threshold. If the value of α_v is less than θ, we consider the molecular emission time prediction for the v-th sample to be correct, assigning $N^{v}_{c} = 1$. Conversely, if the value of α_v is greater than or equal to θ, the prediction is incorrect, setting $N^{v}_{c} = 0$. Therefore, we can calculate the number of samples with correct prediction in the testing dataset as follows:

$$N_c = \sum_{v=1}^{N_1} N^{v}_{c}. \quad (18)$$

We use Ω to represent the accuracy of the emission time under the error time threshold θ. The calculation method involves dividing the number of correctly predicted samples by the total number of samples in the testing dataset as follows:

$$\Omega = \frac{N_c}{N_1}. \quad (19)$$

D. Complexity Analysis

In this paper, the Transformer-based model and the DNN-based model are trained offline. Therefore, we mainly focus on the time complexity of offline training neural networks. In addition, the time complexity of offline training depends on different factors, such as the number of training iterations, the size of the training dataset, the number of neurons in the input layer, the hidden layers and the output layer. The offline time complexity of the Transformer-based model can be represented as $O(L^2 \times \Gamma \times d \times \omega)$ [30]. Here, L represents the length of the input sequence, which corresponds to the sequence of the number of the molecules captured by K receivers in $[0, T]$. Γ is the number of Transformer layers. ω is the number of attention heads, and d denotes the dimension of the hidden layers. For the DNN-based model in this paper, it is assumed that the numbers of neurons in the input layer and hidden layers are A and B, respectively. ξ is the number of training iterations, and δ is the size of the training dataset. According to [31], the time complexity of the DNN-based model can be expressed as $O(\xi \times \delta \times A \times B)$. We can see that since the time complexity of the Transformer-based model is higher than that of the DNN-based model. However, on one hand, we consider these two models are trained offline. On the other hand, the accuracy of prediction by the Transformer-based model is higher than that of the DNN-based model. Therefore, we use the Transformer-based model to predict the emission time and arrival time of the molecules.

IV. Transformer-Based Prediction of Arrival Time

A. Data Generation

The mechanism of estimation of the arrival time is described as follows: for the k-th receiver, Rx_k estimates the arrival time based on the initial coordinates of the molecules which

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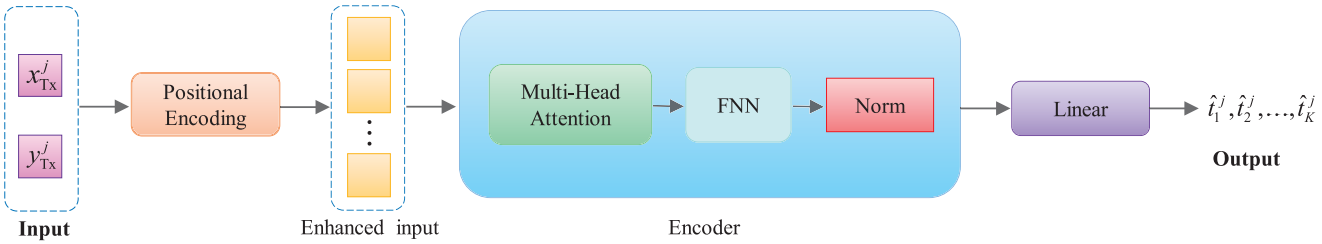


Fig. 3. Transformer-based model for estimating the arrival time.

diffuse in the environment. The estimation of arrival time of Rx_k is defined by the time at which Rx_k captures the first molecule when Tx releases the molecules at t=0. It means that Rx_k determines whether one molecule can be captured in each time step Δt. If no molecules are captured in this Δt, Rx_k continues to count the molecules in subsequent Δt. Once one molecule can be captured at time t, then the time t is the arrival time of Rx_k. When the position of Tx changes, the corresponding arrival time of Rx_k may vary. Therefore, it is essential to employ computer simulations to model the process of emission, diffusion and reception of molecules under different position coordinates of Tx.

To ensure universal applicability of the neural network, the position of Tx in 2D environment needs to be randomly selected. Although the coordinates of Tx can theoretically take continuous values, obtaining continuous coordinate $(x_{\mathrm{Tx}}, y_{\mathrm{Tx}})$ as the training data is a challenge work due to high computational complexity. Therefore, the x-axis and y-axis are discretized into multiple steps and each step is 0.1μm. The ranges of x_{Tx} and y_{Tx} are set the same as formula (6).

When Tx coordinates do not satisfy formula (6), the Tx coordinates are not used for data generation. $(x^j_{\mathrm{Tx}}, y^j_{\mathrm{Tx}})$ is denoted by the coordinate of the corresponding transmitter in the j-th simulation. We use t^j_k to represent the time that the molecule reach the k-th receiver in the j-th simulation, which is the output of the j-th sample, so the representation for the j-th sample W^j_{arrival} in the dataset is as follows:

$$W^j_{\mathrm{arrival}} = \{(x^j_{\mathrm{Tx}}, y^j_{\mathrm{Tx}}), (t^j_1, t^j_2, \ldots, t^j_K)\}. \tag{20}$$

We use I_2 to represent the number of samples in the dataset, then the dataset $F^{I_2}_{\mathrm{arrival}}$ is represented by

$$F^{I_2}_{\mathrm{arrival}} = \left\{ W^1_{\mathrm{arrival}}, W^2_{\mathrm{arrival}}, W^3_{\mathrm{arrival}}, \ldots, W^{I_2}_{\mathrm{arrival}} \right\}. \tag{21}$$

B. Considered Model

On the problem of predicting the arrival time of molecules at the receivers, we use the model in Section III and modify the input and output of the model. As shown in Fig. 3, the input of the model is the j-th sample coordinate $(x^j_{\mathrm{Tx}}, y^j_{\mathrm{Tx}})$ of Tx, and the output is the arrival time of molecules at K receivers $\hat{t}^j_1, \hat{t}^j_2, \ldots, \hat{t}^j_K$. The experimental results show that the Transformer-based model still has a good performance in predicting the time when the first molecule reaches the receivers.

C. Training and Testing Process

Based on the range of x_{Tx} and y_{Tx} in formula (6), there are 160,000 discretized coordinates in 2D environment. For the samples in the dataset, we take 80% as the training dataset and 20% as the testing dataset. We use N_2 to represent the number of samples in the testing dataset and N_2= 32,000. S_2 is a batch size, and $S_2 = 64$. The learning rate is set to 0.001.

We also choose MSE as the loss function in the training process. Assuming that p is the index of the p-th sample in S_2. t^p_k is the real molecular arrival time of the k-th receiver in the p-th sample. $\hat{t}^p_k$ is the molecular arrival time of the k-th receiver predicted by the neural network in the p-th sample. The loss function for estimating the arrival time is calculated as follows:

$$\mathrm{MSE}_{\mathrm{arrival}} = \frac{1}{S_2} \sum_{p=1}^{S_2} \sum_{k=1}^{K} (t^p_k - \hat{t}^p_k)^2. \tag{22}$$

During the testing, we use β^l to represent the error time of the predicted molecular arrival time at K receivers for the l-th sample. The calculation method is as follows:

$$\beta^l = \frac{1}{K}(|t^l_1 - \hat{t}^l_1| + |t^l_2 - \hat{t}^l_2| + \cdots + |t^l_K - \hat{t}^l_K|), \tag{23}$$

where $t^l_1, t^l_2, \ldots, t^l_K$ are the real molecular arrival time of K receivers for the l-th sample. $\hat{t}^l_1, \hat{t}^l_2, \ldots, \hat{t}^l_K$ are the molecular arrival time predicted by the neural network for the l-th sample.

On the whole testing dataset, we use AET to represent the average error time of the entire testing dataset under the error time threshold θ. After obtaining the average error time of each sample, we can calculate AET as

$$\mathrm{AET} = \frac{1}{N_2} \sum_{l=1}^{N_2} \beta^l. \tag{24}$$

V. Simulation Results

A. The Simulation Results of Estimating the Emission Time

After using the method in Section III to obtain the training and testing datasets, we use two deep learning neural networks including the DNN-based model and the Transformer-based model to predict the emission time of molecules, and compare the differences between these two models in solving the problem. The structure of the DNN-based model has 6 hidden layers. The first hidden layer has 200 input units and 2000 output units. Each other hidden layer has 2000 input units and 2000 output units. The learning rate is 0.001. The ReLU

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TABLE I
SIMULATION PARAMETERS FOR ESTIMATING THE EMISSION TIME

Parameter	Variable	Value
Number of receivers	K	2
Radius of Rx_k	r_k	2μm
Coordinates of centers of Rx_1 and Rx_2	$(x_{\mathrm{Rx}_1}, y_{\mathrm{Rx}_1})$, $(x_{\mathrm{Rx}_2}, y_{\mathrm{Rx}_2})$	(-5μm,5μm), (0μm,0μm)
Total time	T	5s
Number of time steps	M	100
Each time step	Δt	0.05s
The latest emission time of molecules	t_{latest}	4s
Number of released molecules	N	5000
Diffusion coefficient of molecules	D	500μm^2/s

TABLE II
ACCURACY COMPARISON RESULTS BETWEEN TWO MODELS

Model	Transformer-based	DNN-based
Ω ($\theta = 1\Delta t$)	54.13%	15.76%
Ω ($\theta = 2\Delta t$)	84.34%	31.58%
Ω ($\theta = 3\Delta t$)	95.37%	45.93%

activation function is added between the hidden layers to accelerate the convergence speed during the training process. Finally, we use the linear layer transformation to output the emission time of molecules. The parameters of the simulation experiment for estimating the emission time of molecules are given in Table I.

Through obtaining the experimental results for both the Transformer-based and DNN-based models, we offer a concise statistical analysis in Table II.

In Table II, it is shown that when the error time threshold θ is set as $1\Delta t$, $2\Delta t$ and $3\Delta t$, the prediction performance of the Transformer-based model is compared with that of the DNN-based model. When $\theta = 1\Delta t$, the Transformer-based model achieves the accuracy of over 50% on the entire testing dataset, while the DNN-based model only reaches 15.76% . For $\theta = 2\Delta t$ and $\theta = 3\Delta t$, the accuracy of prediction of the Transformer-based model is significantly higher than that of the DNN-based model.

The $\mathrm{MSE}_{\mathrm{emission}}$ convergence of the loss function during the training of the Transformer-based model and the DNN-based model is shown in Fig. 4. Obviously, when the Transformer-based model and the DNN-based model are trained at the beginning, the value of $\mathrm{MSE}_{\mathrm{emission}}$ decreases very fast. But when the value of the epoch increases, the value of the loss function of the Transformer-based model is always smaller than that of the DNN-based model, which indicates that the Transformer-based model has better performance. When the value of the epoch reaches 40, the Transformer-based model converges, and the convergence value is much smaller than that of the DNN-based model.

There are some reasons why we choose 128,400 samples with the allocation ratio of 20%. In order to show the impacts of data distribution characteristics on model performance, we adjust the number of samples in the dataset, and 20% of the samples in the dataset are allocated to the testing dataset.

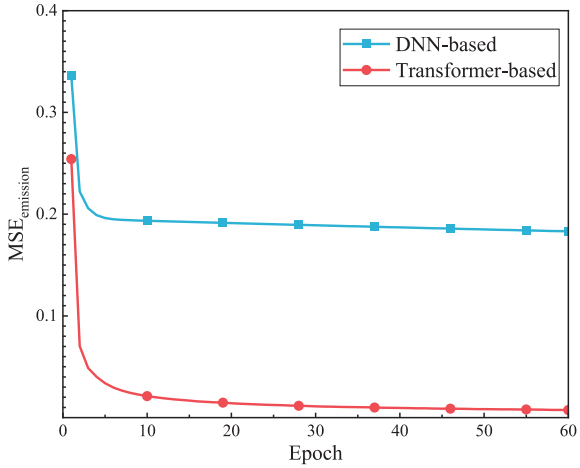


Fig. 4. The comparison results of convergence between two models when estimating the emission time.

TABLE III
THE ACCURACY OF THE EMISSION TIME VS THE NUMBER OF SAMPLES IN THE DATASET

Model	Ψ =83,200	Ψ =124,800	Ψ =166,400
Ω ($\theta = 1\Delta t$)	50.11%	54.13%	55.03%
Ω ($\theta = 2\Delta t$)	81.74%	84.34%	84.52%
Ω ($\theta = 3\Delta t$)	93.47%	95.37%	95.56%

TABLE IV
THE ACCURACY OF THE EMISSION TIME VS THE ALLOCATION RATIO OF THE TESTING DATASET

Model	$R = 10\%$	$R = 20\%$	$R = 30\%$
Ω ($\theta = 1\Delta t$)	54.61%	54.13%	53.97%
Ω ($\theta = 2\Delta t$)	84.54%	84.34%	84.32%
Ω ($\theta = 3\Delta t$)	95.41%	95.37%	95.26%

We assume that Ψ denotes the number of samples in the dataset. Now we use the same data generation method to generate Ψ=83,200 and Ψ=166,400 data samples, respectively. Then we use these three different datasets to evaluate the performance of the Transformer-based model. The results are shown in Table III.

It can be seen from Table III that increasing the number of samples in the dataset can improve the prediction accuracy of the model. When $\theta = 1\Delta t$, the accuracy results are 50.11%, 54.13%, and 55.03%, respectively. When $\theta = 2\Delta t$, the accuracy under Ψ=166,400 is 84.52%, which is larger than that under Ψ=124,800, which is 84.34%. On one hand, the performance improvement of the model is no longer obvious when increasing the number of samples in the dataset to 166,400. On the other hand, it will increase the complexity of the training and slow down the convergence speed of the model. Considering the balance of the performance of the model and the training complexity, we choose 128,400 samples when estimating the emission time of the molecules.

We then adjust the allocation ratio R of the testing dataset when fixing the number of samples in the dataset. We take $R = 10\%$, $R = 20\%$ and $R = 30\%$, respectively. The accuracy results of the model prediction are shown in Table IV. According to Table IV, we can see that decreasing R can improve the accuracy of the model prediction. When $\theta = 1\Delta t$,

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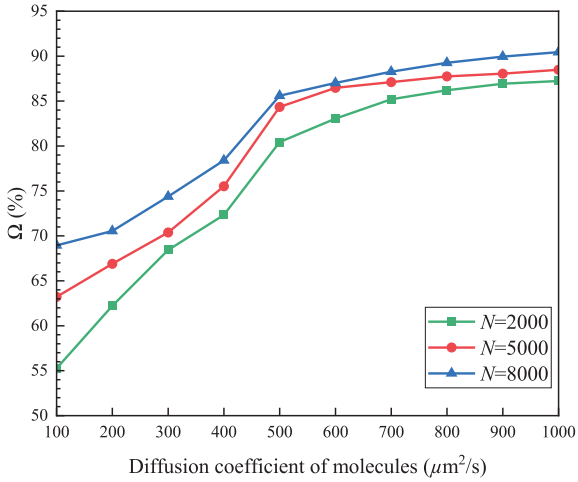


Fig. 5. The accuracy with different diffusion coefficients of molecules ($\theta = 2\Delta t$).

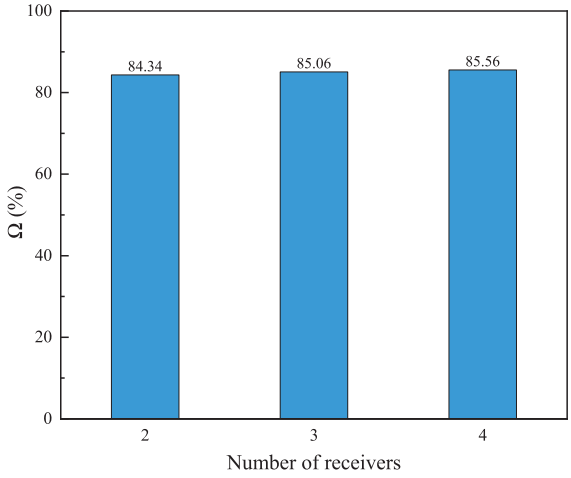


Fig. 6. The accuracy with different numbers of receivers ($\theta = 2\Delta t$).

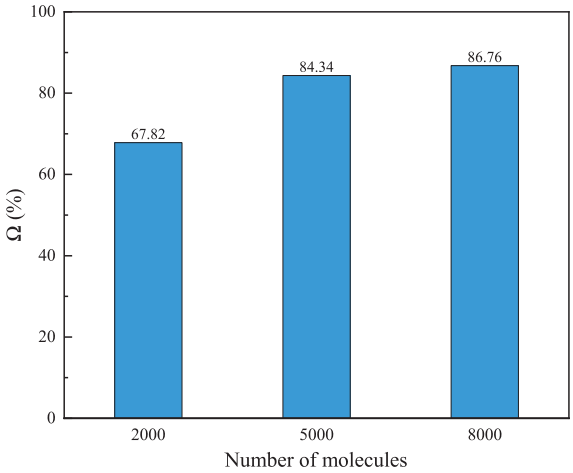


Fig. 7. The accuracy with different numbers of molecules ($\theta = 2\Delta t$).

the accuracy results are 54.61%, 54.13%, and 53.97%, respectively. When $\theta = 2\Delta t$, the accuracy is 84.54% under $R = 10\%$ which is larger than that under $R = 20\%$. When $R = 20\%$ and then we reduce the value of R, the prediction accuracy of the model is no longer obvious. Then we choose the allocation ratio $R = 20\%$ to obtain the simulation results.

When the numbers of released molecules N are 2000, 5000 and 8000, respectively, we take the diffusion coefficients of the molecules as $100\mu\text{m}^2/\text{s}$, $200\mu\text{m}^2/\text{s}$, ..., $900\mu\text{m}^2/\text{s}$ and $1000\mu\text{m}^2/\text{s}$, respectively. The three curves obtained are shown in Fig. 5. Firstly, we can see that when N becomes larger and the accuracy of the model prediction also becomes higher. Secondly, the trend of the curves clearly indicates that an increase of diffusion coefficient of molecules enhances the accuracy of prediction. This can be attributed to the fact that higher value of diffusion coefficient allows molecules to reach the receiver more quickly, resulting in a greater capture rate and thus improving the model's ability to predict the emission time of molecules. Notably, within the diffusion coefficient range of $[500, 1000]\mu\text{m}^2/\text{s}$, the increase of accuracy of prediction occurs at a slower rate compared to the range of $[100, 500]\mu\text{m}^2/\text{s}$. This observation suggests that once the diffusion coefficient reaches $500\mu\text{m}^2/\text{s}$, further increase may not significantly enhance the accuracy, implying that other parameters may also influence the prediction results. In addition, when the value of the diffusion coefficient is fixed, the larger value of N, the higher value of the accuracy. This result demonstrates that more released molecules can achieve more precise emission time.

Fig. 6 illustrates the influence of the number of receivers on the accuracy of model predictions. It is evident that when the number of receivers is 2, the accuracy of predicting the emission time of molecules by the Transformer-based model is 84.34%. With 3 receivers, the accuracy slightly improves to 85.06%, and with 4 receivers, it reaches 85.56%. When $K > 2$, the accuracy of the model does not show significant improvement, which remains around 85%. Because the number of molecules released by Tx is constant, when there are multiple receivers with $K = 2$, increasing the number of receivers will reduce the number of molecules captured by each receiver in $[0, T]$. The characteristics of the sequences captured by the receivers does not significantly increase, so the accuracy of prediction is not greatly improved. From this result, we can see that two receivers already provide satisfactory prediction accuracy. Therefore, we ultimately choose $K=2$ to decrease the resource consumption for obtaining the simulation results.

Fig. 7 shows the effect of the number of emitted molecules on the predicted molecular emission time by this model. When the number of molecules is 2000, the prediction accuracy of the model is only 67.82%. When the number of molecules is 5000, the prediction accuracy of the model is 84.34%. It can be seen that the increase of the number of molecules can improve the prediction accuracy of the model. The reason for this phenomenon is that when the number of released molecules increases, the number of molecules that can be captured by each receiver in each Δt also increases. The increase of the number of captured molecules is beneficial to the model to predict the emission time of the molecules. When the number of molecules is 8000, the prediction accuracy of the model is 86.76%. In contrast, the accuracy is not significantly improved. The reason is that when the number of molecules is sufficient,

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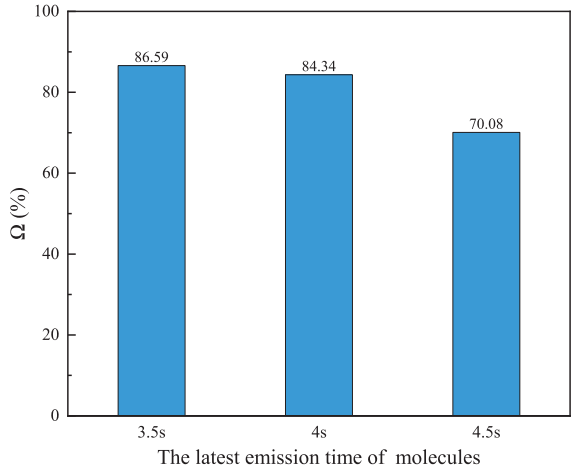


Fig. 8. The accuracy with different latest emission time of molecules ($\theta = 2\Delta t$).

other factors including the diffusion coefficient of molecules, the number of receivers, the latest emission time of molecules will also affect the accuracy of prediction.

Fig. 8 illustrates the influence of selecting different latest molecular emission time in [0, 5]s. When the latest molecular emission time is 3.5s, the prediction accuracy of the model is 86.76%. The reason for this phenomenon is that the emitted molecules have enough time to reach the receiver, and the receiver captures enough molecules to help the model to predict more accurately. When the latest molecular emission time is 4.5s, the prediction accuracy of the model is 70.08%. The reason is that after the molecules are released, the receiver only has 0.5s time to capture the molecules. Then the molecules need enough time to diffuse to arrive at the receiver. It will lead to a large number of molecules which can not reach the receiver in time. Thus the number of molecules captured by the receiver will decrease.

TABLE V
SIMULATION PARAMETERS FOR ESTIMATING THE ARRIVAL TIME

Parameter	Variable	Value
Number of receivers	K	3
Radius of Rx_k	r_k	2μm
Coordinates of centers of Rx_1, Rx_2 and Rx_3	$(x_{\mathrm{Rx}_1}, y_{\mathrm{Rx}_1})$, $(x_{\mathrm{Rx}_2}, y_{\mathrm{Rx}_2})$ and $(x_{\mathrm{Rx}_3}, y_{\mathrm{Rx}_3})$	(-5μm,5μm), (0μm,0μm) and (6μm,5μm)
Each time step	Δt	0.01s
Number of molecules released	N	5000
Diffusion coefficient of molecules	D	500$\mu\mathrm{m}^2$/s

TABLE VI
AET COMPARISON RESULTS BETWEEN TWO MODELS

Model	Transformer-based	DNN-based
AET ($N = 2000$)	$2.21\Delta t$	$2.61\Delta t$
AET ($N = 5000$)	$1.27\Delta t$	$2.13\Delta t$
AET ($N = 8000$)	$0.95\Delta t$	$1.47\Delta t$

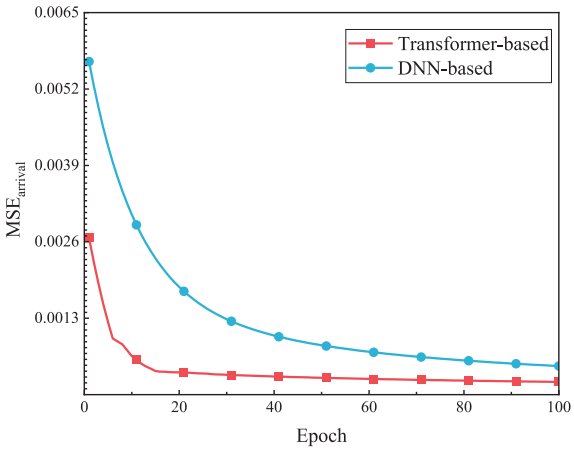


Fig. 9. The comparison results of convergence between two models when estimating the arrival time.

B. The Simulation Results of Estimating the Arrival Time

Similar to the previous section, after obtaining the dataset of molecular arrival time, we train and test the DNN-based model and Transformer-based model, respectively. The structure of the DNN-based model has 6 hidden layers. The first hidden layer has 2 input units and 200 output units. Each other hidden layer has 200 input units and 3 output units. The learning rate is 0.001. The ReLU activation function is added between the hidden layers and we use the linear layer transformation to output the arrival time of molecules. The parameters of the simulation experiment for estimating the arrival time are given in Table V.

Table VI gives the comparison results of the average error time AET of Transformer-based model and DNN-based model in the testing dataset. The calculation method of AET has been introduced in Eq. (24). When the number of released molecules N are 2000, 5000 and 8000, respectively, it can be seen that the AET of Transformer-based model is smaller than that of DNN-based model. When $N = 5000$, the AET of the Transformer-based model is $1.27\Delta t$, and the AET of the DNN-based model is $2.13\Delta t$. At this time, the performance difference between the two models is the largest.

Fig. 9 reveals the changes of the loss function of the Transformer-based model and the DNN-based model on the training dataset. We can see that the initial loss value of the DNN-based model is larger than that of the Transformer-based model. In terms of convergence speed, the Transformer-based model converges at about the $60th$ epoch, while the DNN-based model converges at about the $90th$ epoch. When the loss function reaches convergence, the $\mathrm{MSE}_{\mathrm{arrival}}$ corresponding to the Transformer-based model is also smaller than that of the DNN-based model. Results show that the Transformer-based model has better performance.

Fig. 10 evaluates the influence of different number of molecules on the arrival time predicted by the receiver. When the diffusion coefficients of the molecules are $200\mu\mathrm{m}^2/\mathrm{s}$, $500\mu\mathrm{m}^2/\mathrm{s}$, $800\mu\mathrm{m}^2/\mathrm{s}$, respectively. We take the number of molecules as $1000, 2000, \ldots, 9000, 10000$, respectively. It can be seen from the height of the three curves that the

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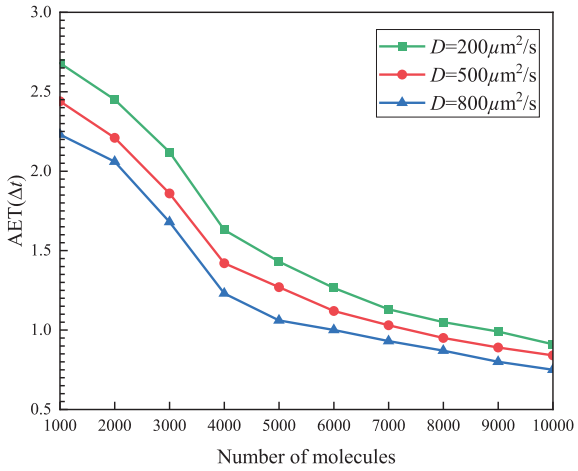


Fig. 10. The AET with different numbers of molecules.

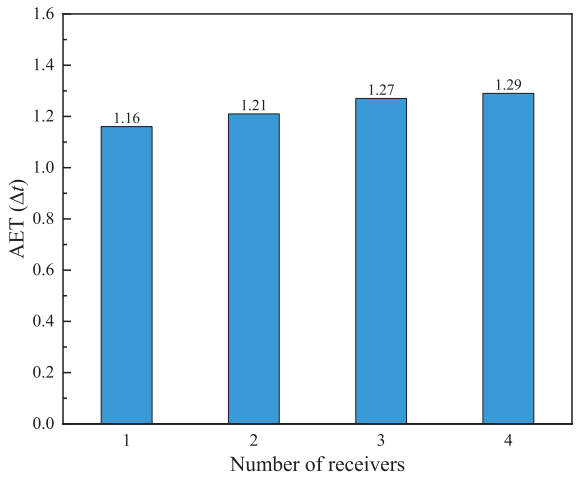


Fig. 11. The AET with different numbers of receivers.

diffusion coefficient of the molecule becomes larger and the average error time AET of the testing dataset becomes smaller. Secondly, it can be seen from the downward trend of the curve that as the number of emitted molecules increases, the AET decreases. Because as the number of molecules in the environment increases, the variation for capturing the first molecule by the receivers is small, which makes the training of our model more effective. Thirdly, as the value of N increases, the number of captured molecules in the received sequence will increase significantly, and the AET will decrease significantly. When the value of N reaches 4000, the magnitude of the increase in the number of received molecules is smaller than that under the case $N < 4000$. Then the data characteristics do not change significantly, so the AET decreases slowly.

Predicting the arrival time of molecules at multiple receivers can better demonstrate the good prediction ability of the Transformer-based model. As shown in Fig. 11, we experiment with 1, 2, 3 and 4 receivers respectively. When the number of receivers is 1, the prediction AET of the model is $1.16\Delta t$. When the number of receivers increases, the prediction AET of the model increases slightly. With the

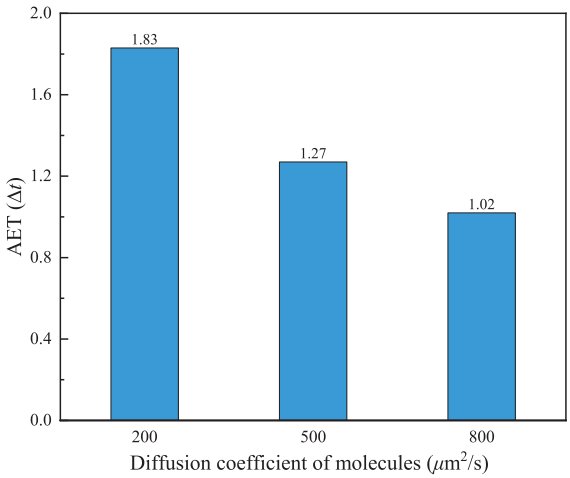


Fig. 12. The AET with different diffusion coefficients of molecules.

increasing value of K, the accuracy of predicting the emission time of molecules is increasing. When K=4, the accuracy is higher than that under K=3. However, the resource consumption, such as the time complexity under K=4 is also higher than that under K=3. In order to balance the accuracy of predicting the emission time of molecules and the time complexity of the model, we choose K=3.

In Fig. 12, we take the diffusion coefficients of the molecules as $200\mu\mathrm{m}^2/\mathrm{s}$, $500\mu\mathrm{m}^2/\mathrm{s}$ and $800\mu\mathrm{m}^2/\mathrm{s}$, respectively. When the diffusion coefficient of the molecules is $200\mu\mathrm{m}^2/\mathrm{s}$, the AET of the model is $1.83\Delta t$. When the diffusion coefficient of the molecules is $500\mu\mathrm{m}^2/\mathrm{s}$, the AET of the model is $1.27\Delta t$. It can be found that the AET decreases by $0.56\Delta t$. When the diffusion coefficient of the molecules is $800\mu\mathrm{m}^2/\mathrm{s}$, the AET of the model is $1.02\Delta t$. Compared with the diffusion coefficient of $500\mu\mathrm{m}^2/\mathrm{s}$, it can be found that the AET is not significantly reduced. The reason for the above phenomenon is that the larger diffusion coefficient can make the molecules reach the receiver faster. Then the number of molecules captured by the receiver in a fixed time will increase, which is more conducive to the prediction of the model. However, when the diffusion coefficient increases to a certain value, it will no longer significantly affect the prediction accuracy of the arrival time of the model.

In our work, we have proposed the method which utilizes the Transformer model to estimate the emission time and arrival time of molecules in DMC system. Now we analyze the feasibility of the proposed method in DMC channels. Deep learning algorithm is run on a macro-scale device. In this paper, we assume the link between the receivers and the external device is reliable. Therefore, we study the case of the ideal communication channel between the receivers and the external device. On one hand, the reliability of the communication link is also the direction of our future research. On the other hand, this assumption is based on the following references. Torres Gómez et al. [18] proposed the explainability of deep learning for symbol detection in DMC channels by using real testbed measurements. Khaloopour et al. [32] studied the

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sensing, communication, and localization in DMC by considered the ideal (noiseless) or nonideal (noisy) communication channel between the fusion center and the gateway. Based on above analysis, the proposed method has the feasibility in DMC.

VI. Conclusion

This paper proposed a Transformer-based model to predict the emission time of and the arrival time of molecules in the case of unknown CSI, respectively. The model structures used in the two tasks are similar. However, the input and output of the Transformer-based model are different when solving these two problems. This method not only has good performance in 2D unknown CSI environment, but also can be easily extended to 3D unknown CSI environment. To validate the superiority of the model, we conducted comparative experiments with a DNN-based model. The results demonstrated that our model achieved better performance. Through simulation experiments, we found the factors that affected the prediction of emission time and arrival time of molecules, including the number of receivers, the number of released molecules, and the diffusion coefficient of molecules. When the number of receivers, the number of released molecules and the diffusion coefficient take larger values in these two tasks by this model will be better. In particular, the latest release time of the molecules will also affect the model to predict the emission time. When the latest release time of the molecules becomes smaller, the accuracy of the model prediction becomes higher.

There are also some potential challenges of the proposed method in actual DMC system. First, we need to perceive the channel between a receiver and a macro device and the link between the receiver and the external device must be extremely reliable. Second, the complexity of the Transformer-based model should be decreased and optimized to adapt to practical application scenarios in DMC system. Furthermore, We plan to use deep learning methods to predict the emission time and the arrival time of molecules in the MMC scenario with mobile transmitters and mobile receivers.

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